

Session 4:

Software Development Lifecycle (SDLC) Explain SDLC phases: Requirement Design Development Testing Deployment Maintenance. Discuss Agile vs Waterfall, Scrum roles (PO, Scrum Master, Dev Team). Relate each phase to a real-world app (e.g., “Spotify new feature rollout”). Trainer Demo: Show a Jira board mock-up with stories in “To Do / In Progress / Done”

1) LIST ALL 6 PHASES OF SDLC IN ORDER WITH ONE LINE ABOUT EACH

PHASE	DESCRIPTION
1. REQUIREMENT ANALYSIS	GATHER AND UNDERSTAND THE PROJECT REQUIREMENTS FROM STAKEHOLDERS.
2. DESIGN	CREATE THE SYSTEM ARCHITECTURE, DATABASE DESIGN, AND UI DESIGN.
3. DEVELOPMENT	WRITE AND BUILD THE APPLICATION CODE.
4. TESTING	VERIFY THAT THE SOFTWARE WORKS CORRECTLY AND FIX DEFECTS.
5. DEPLOYMENT	RELEASE THE SOFTWARE TO USERS OR THE PRODUCTION ENVIRONMENT.
6. MAINTENANCE	PROVIDE UPDATES, BUG FIXES, AND ENHANCEMENTS AFTER RELEASE.

2) WRITE 3 DIFFERENCES BETWEEN AGILE AND WATERFALL MODELS

AGILE	WATERFALL
DEVELOPMENT IS ITERATIVE AND INCREMENTAL.	DEVELOPMENT FOLLOWS A FIXED SEQUENTIAL PROCESS.

AGILE

REQUIREMENTS CAN CHANGE DURING THE PROJECT.

FREQUENT RELEASES AND CUSTOMER FEEDBACK.

WATERFALL

REQUIREMENTS ARE DEFINED AT THE BEGINNING AND RARELY CHANGE.

PRODUCT IS USUALLY DELIVERED AT THE END OF THE PROJECT.

3) NAME THE 3 MAIN SCRUM ROLES AND ONE RESPONSIBILITY OF EACH

ROLE	RESPONSIBILITY
PRODUCT OWNER (PO)	DEFINES AND PRIORITIZES PRODUCT REQUIREMENTS (BACKLOG).
SCRUM MASTER	FACILITATES SCRUM PRACTICES AND REMOVES OBSTACLES FOR THE TEAM.
DEVELOPMENT TEAM	DESIGNS, DEVELOPS, AND TESTS THE PRODUCT INCREMENT.

4) WHAT IS A SPRINT IN SCRUM? MENTION ITS TYPICAL DURATION.

A SPRINT IS A FIXED TIME PERIOD DURING WHICH THE SCRUM TEAM COMPLETES A SET OF PLANNED WORK ITEMS AND DELIVERS A USABLE PRODUCT INCREMENT.

TYPICAL DURATION: 1 TO 4 WEEKS (COMMONLY 2 WEEKS).

5) WHAT DO THESE JIRA BOARD COLUMNS MEAN?

COLUMN	MEANING
BACKLOG	LIST OF TASKS, FEATURES, AND BUGS WAITING TO BE WORKED ON.
TO DO	TASKS SELECTED AND READY TO START.
IN PROGRESS	TASKS CURRENTLY BEING WORKED ON BY THE TEAM.
DONE	TASKS THAT HAVE BEEN COMPLETED AND APPROVED.